

Predictive Model Plan

1. Model Logic (Generated with GenAI)

1.1 Model Purpose

The predictive delinquency model is designed to identify customers who are at risk of missing future payments by analyzing their demographic information, financial characteristics, and historical payment behavior. By learning patterns associated with account delinquency (Delinquent_Account), the model estimates the probability that a customer will become delinquent and classifies customers into operational risk categories (High, Medium, or Low Risk). These predictions enable Geldium's Collections team to prioritize high-risk customers and implement proactive intervention strategies to reduce delinquency rates.

1.2 Predictive Model Logic (Pseudocode)

GELDIUM DELINQUENCY PREDICTION ENGINE LOGIC WORKFLOW

INPUT:

Customer dataset containing demographic information, financial metrics, and Month_1 to Month_6 payment history.

STEP 1: LOAD CUSTOMER DATA

- Import customer demographic, financial, and payment history data.

STEP 2: PREPROCESS DATA

- Impute missing numerical values using median imputation.
- Impute missing categorical values using mode imputation.
- Standardize inconsistent categorical values (e.g., EMP → Employed).
- Encode categorical variables using One-Hot Encoding.
- Perform additional preprocessing if required.

STEP 3: FEATURE ENGINEERING & SELECTION

- Convert payment status into numerical values:

On-time = 0

Late = 1

Missed = 2

- Generate additional behavioral features:

- Count_Missed_Months
- Count_Late_Months

- Select important predictors:

- Credit Utilization
- Credit Score
- Debt-to-Income Ratio
- Income
- Loan Balance
- Missed Payments
- Payment History (Month_1–Month_6)
- Employment Status
- Account Tenure

STEP 4: SPLIT DATASET

- Divide the dataset into:

- Training Set (80%)
- Testing Set (20%)

- Apply stratified sampling to preserve the delinquency class distribution.

STEP 5: TRAIN RANDOM FOREST MODEL

- Train a Random Forest classifier using the training dataset.

- Apply balanced class weights to reduce the impact of class imbalance.

STEP 6: GENERATE PREDICTIONS

- Predict the probability of customer delinquency.

- Classify customers into:

- High Risk
- Medium Risk

- Low Risk

STEP 7: OUTPUT RESULTS

- Generate:

- Probability of delinquency
- Predicted class (Delinquent / Not Delinquent)
- Risk category
- Feature importance rankings

STEP 8: EVALUATE MODEL PERFORMANCE

- Evaluate the model using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

OUTPUT:

A predicted delinquency probability, customer risk category, and feature importance scores to support the Collections team's decision-making.

1.3 Chosen Model Summary

We selected a Random Forest Classifier because it effectively captures complex, non-linear relationships within structured credit data without making strict distribution assumptions. After appropriate preprocessing and encoding of categorical variables, it can model interactions between financial and behavioral features while reducing the risk of overfitting through ensemble learning. Additionally, it provides feature importance rankings, improving explainability and supporting transparent risk assessment.

1.4 Top 5 Input Features

Input Feature	Importance to Prediction
Payment History (Month_1–Month_6)	Historical payment behavior is one of the strongest indicators of future delinquency.
Credit Utilization	High credit utilization may indicate financial stress and increased repayment risk.
Credit Score	Reflects a customer's creditworthiness and overall repayment history.
Debt-to-Income Ratio	A higher debt burden relative to income may reduce repayment capacity.
Missed Payments	Previous missed payments may indicate an increased likelihood of future delinquency.

2. Justification for Model Choice

A Random Forest Classifier was selected because it provides a good balance between predictive performance, interpretability, and operational practicality for structured financial data. Compared with Logistic Regression, which assumes linear relationships between variables, Random Forest can capture more complex interactions among financial indicators and payment behavior without requiring those assumptions. Compared with a single Decision Tree, it reduces the risk of overfitting by combining the predictions of multiple trees, resulting in more reliable performance on unseen data.

From a transparency perspective, Random Forest provides feature importance scores, allowing Geldium to identify which variables contribute most to delinquency predictions. This supports explainable, data-driven decision-making and helps business stakeholders understand the factors influencing customer risk.

The model is also practical to implement because it handles both numerical and categorical variables effectively after preprocessing and requires relatively little feature engineering. Additionally, class imbalance can be addressed using balanced class weights during training, helping the model identify delinquent customers more effectively without being biased toward the majority class.

Overall, Random Forest aligns well with Geldium's business objectives by producing reliable

probability scores that support customer risk segmentation and enable the Collections team to prioritize proactive interventions. It combines strong predictive capability with sufficient explainability, making it an appropriate choice for AI-assisted credit risk assessment.

Action: Summary Justification Paragraph

A Random Forest Classifier was selected for Geldium's delinquency prediction because it provides a strong balance between predictive performance, explainability, and operational practicality. It can capture complex relationships among customer financial characteristics and payment history while providing feature importance scores that help explain model predictions. By supporting class imbalance handling and generating probability-based risk scores, the model enables Geldium's Collections team to identify high-risk customers, prioritize intervention strategies, and make transparent, data-driven decisions.

3. Evaluation Strategy

To evaluate the performance of the proposed Random Forest model, a combination of performance metrics, fairness checks, and ethical considerations will be used. Since only 16% of the customers in Geldium's dataset are delinquent, relying solely on accuracy may give misleading results. Therefore, multiple evaluation metrics will be used to assess both prediction quality and model reliability.

3.1 Performance Metrics

Metric	Purpose	Interpretation
Accuracy	Measures the percentage of correctly classified customers.	Provides an overall measure of performance but should not be used alone because the dataset is imbalanced (84% non-delinquent, 16% delinquent).
Precision	Measures how many customers predicted as delinquent are actually delinquent.	High precision reduces unnecessary collection efforts caused by false positives.
Recall (Sensitivity)	Measures how many actual delinquent customers are correctly identified.	High recall is especially important because failing to identify high-risk customers could result in increased financial losses.
F1-Score	Combines precision and recall into a single metric.	A higher F1-score indicates a good balance between identifying delinquent customers and minimizing false alarms.

ROC-AUC	Evaluates how well the model distinguishes between delinquent and non-delinquent customers across different classification thresholds.	Higher ROC-AUC values indicate better overall discrimination ability.
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3.2 Model Reliability

To ensure reliable predictions, the dataset will be divided into 80% training data and 20% testing data using stratified sampling so that both sets maintain the original delinquency distribution (84% non-delinquent and 16% delinquent). The model's performance will be evaluated on the testing dataset, and cross-validation may be used to verify that the results remain consistent across different data splits.

3.3 Bias Detection and Fairness

To promote fair and responsible AI, the model's performance will be evaluated across different customer groups, including age groups, employment status, and location, to identify any significant differences in prediction accuracy or error rates. If bias is detected, preprocessing techniques, balanced class weights, or feature review can be applied to reduce unfair outcomes while maintaining predictive performance. Sensitive demographic attributes should not be used as the sole basis for predicting customer risk.

3.4 Ethical Considerations

The model should support decision-making rather than replace human judgment. Predictions should be used to help the Collections team prioritize customer outreach and provide proactive support instead of making fully automated financial decisions. Customer data should be handled securely, prediction results should be explainable, and the model should be monitored regularly to ensure it remains accurate, fair, and aligned with responsible AI practices.

Action: Evaluation Plan Summary

The proposed evaluation strategy combines Accuracy, Precision, Recall, F1-Score, and ROC-AUC to measure the model's predictive performance comprehensively. Because

the dataset is imbalanced, greater emphasis will be placed on Recall and F1-Score to ensure high-risk customers are identified while maintaining a reasonable balance between false positives and false negatives. Fairness checks will be performed across different customer groups, and ethical AI principles will be followed to ensure transparent, reliable, and responsible delinquency prediction.